Arnav Mahishi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.02
Sri Kumarans Childrens Home	Class XII (CBSE)	2024	96.20%
Greenwood High International School	Class X (ICSE)	2022	97.00%

WORK EXPERIENCE

AI Intern | Simpli-AI | Bangalore, India

- Built a multilingual call grading system that evaluates customer support conversations and grades parameters across coherence, context retention, resolution quality, sentiment trajectory, and professionalism using transformer-based NLP models.
- Developed a live gender detection module within the call processing pipeline to enrich real-time caller analytics.
- Designed and trained a logistic regression resolution classifier using text embeddings to automatically determine the resolution outcome of the call.
- Built the company website, improving UI/UX and overall product presentation. (Link)

AI Intern | Simpli-AI | Bangalore, India

- Evaluated enterprise conversational AI systems by identifying hallucinations, response failures, and logical inconsistencies.
- Analyzed large-scale chatbot conversation logs using Python to detect intent errors, latency issues, and user experience bottlenecks.
- Investigated failed customer interactions and recommended improvements across ASR, NLU, and response generation pipelines

SKILLS

- **Programming Languages**-Python, C, Verilog, SQL, Linux Shell Scripting, LaTeX
- **AI/ML**-PyTorch, Scikit-learn, YOLOv8, NumPy, Pandas, Matplotlib, Machine Learning
- **Hardware & Embedded**-Arduino, Digital Electronics, Flip-Flops, LCD, 7-Segment Displays
- **Tools**-Git, Docker, Linux, Spike RISC-V Simulator
- **Concepts**- Agentic AI, Data Structures & Algorithms, Machine Learning, Signal Processing, Digital Design, Computer Architecture

SCHOLASTIC ACHIEVMENTS

- 2nd Place - Dream 11 Gameathon 1.0 conducted by **Federation of Indian Fantasy Sports** (INR 7.5 lakh prize)
- **Codeforces** Rating 1830 (Peak)
- **Achieved AIR 159** in **ICPC**, securing **6th position** among participating teams from the college.
- **Optiver Tradeathon**- Ranked 27th overall at the Optiver Tradeathon, a quantitative trading simulation competition.

RELEVANT COURSES

- **Computer Science & AI**: Machine Learning, Computer Architecture, Introduction to Programming(C Programming), Data Structures and Applications

- **Mathematics**: Probability, Random Processes, Statistics, Linear Algebra, Matrix Theory, Numerical Methods, Differential Equations and Transform Techniques

- **Electrical & Electronics**: Digital Signal Processing, Digital Systems, Semiconductor Devices, Circuits and Network Analysis, Engineering Electromagnetics

PROJECTS

AI-Driven Cricket Player Performance Prediction | *Dream11 Gameathon 1.0* | Link

- Developed a Random Forest regressor to predict fantasy cricket player performance using historical match statistics.
- Designed a custom fantasy scoring metric by combining batting, bowling, and consistency-based performance indicators.
- Built an automated pipeline for data preprocessing, feature engineering, fantasy score prediction, and team generation.
- Used Docker to deploy a portable app that automatically generates the optimal Dream11 team.

Multilingual Call Grading System | Link

- An AI-powered pipeline that scores customer support call transcripts (audio + text)
- Evaluates coherence, context retention, resolution quality, sentiment trajectory, professionalism, and audio quality
- Produces a final experience score for each call
- Works across English, Hindi, Arabic, Spanish, French, and other languages using multilingual transformer embeddings

FFT Acceleration on RISC-V | *Prof . Rajesh Kedia* | Link

- Designed custom RISC-V ISA extensions to accelerate FFT using domain-specific hardware instructions.
- Implemented a fixed-point Q31 FFT with CORDIC-based twiddle factor generation and saturating arithmetic.
- Optimized radix-2 butterfly operations using packed complex arithmetic, hardware bit reversal, and custom instructions.
- Reduced FFT latency and instruction count while maintaining high accuracy without floating-point hardware.

Efficient Eigenvalue Computation | Link

- Implemented the QR Algorithm with Wilkinson's shift to compute eigenvalues and eigenvectors of complex matrices.

POSITIONS OF RESPONSIBILITY

Core Member - Machine Learning Domain, Tinkerer's Lab, IIT Hyderabad

EXTRACURRICULAR

Represented IIT Hyderabad in Inter IIT Sports Meet (Basketball)